บรรณานุกรม

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[3] ภาษา VHDL สำหรับการออกแบบวงจรดิจิตอล, ชำนาญ ปัญญาใส วัชรากร หนูทอง กรุงเทพ, ซีเอ็ดยูเคชัน, 2547

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ภาคผนวก ก.

จำนวนและตำแหน่งของแท็ปของอัลกอริทึม LFSR

ตารางที่ ก.1 แสดงช่วงความยาวของรอบและตำแหน่งของแท็ปเมื่อจำนวนบิตเปลี่ยนไป

จำนวนบิต	ช่วงความยาวของรอบ	ตำแหน่งแท็ป(แบบ Galois)
2	3	0,1
3	7	0,2
4	15	0,3
5	31	1,4
6	63	0,5
7	127	0,6
8	255	1,2,3,7
9	511	3,8
10	1023	2,9
11	2047	1,10
12	4095	0,3,5,11
13	8191	0,2,3,12
14	16383	0,2,4,13
15	32767	0,14
16	65535	1,2,4,15
17	131071	2,16
18	262143	6,17
19	524287	0,1,4,18
20	1048575	2,19
21	2097151	1,20
22	4194303	0,21
23	8388607	4,22
24	16777215	0,2,3,23
25	33554431	2,24

ตารางที่ ก.1 (ต่อ)

26	67108863	0,1,5,25
27	134217727	0,1,4,26
28	268435455	2,27
29	536870911	1,28
30	1073741823	0,3,5,29
31	2147483647	2,30
32	4294967295	1,5,6,31
33	4294967295	12,32
34	17179869183	33,32,31,6
35	34359738367	34,1
36	68719476735	35,2
37	1.37439E+11	36,35,34,33,32,31
38	2.74878E+11	37,36,32,31
39	5.49756E+11	38,3
40	1.09951E+12	39,20,18,1
41	2.19902E+12	40,2
42	4.39805E+12	41,22,21,0
43	8.79609E+12	42,5,4,0
44	1.75922E+13	43,26,25,0
45	3.51844E+13	44,3,2,0
46	7.03687E+13	45,20,19,0
47	1.40737E+14	46,4
48	2.81475E+14	47,27,26,0
49	5.6295E+14	48,8
50	1.1259E+15	49,26,25,0
51	2.2518E+15	50,15,14,0
52	4.5036E+15	51,2
53	9.0072E+15	52,15,14,0
54	1.80144E+16	53,36,35,0
55	3.60288E+16	54,23

ตารางที่ ก.1 (ต่อ)

56	7.20576E+16	55,21,20,0
57	1.44115E+17	56,6
58	2.8823E+17	57,18
59	5.76461E+17	58,21,20,0
60	1.15292E+18	59,0
61	2.30584E+18	60,15,14,0
62	4.61169E+18	61,56,55,0
63	9.22337E+18	62,0
64	1.84467E+19	63,3,2,0
65	3.68935E+19	64,17
66	7.3787E+19	65,9,8,0
67	1.47574E+20	66,9,8,0
68	2.95148E+20	67,8
69	5.90296E+20	68,28,26,1
70	1.18059E+21	69,15,14,0
71	2.36118E+21	70,5
72	4.72237E+21	71,5246,5
73	9.44473E+21	72,24
74	1.88895E+22	73,15,14,0
75	3.77789E+22	74,10,9,0
76	7.55579E+22	75,35,34,0
77	1.51116E+23	76,3029,0
78	3.02231E+23	77,19,18,0
79	6.04463E+23	78,8
80	1.20893E+24	79,37,36,0
81	2.41785E+24	80,3
82	4.8357E+24	81,37,34,2
83	9.67141E+24	82,45,44,0
84	1.93428E+25	83,12
85	3.86856E+25	84,27,26,0

ตารางที่ ก.1 (ต่อ)

86	7.73713E+25	85,12,11,0
87	1.54743E+26	86,12
88	3.09485E+26	87,71,70,0
89	6.1897E+26	88,37
90	1.23794E+27	89,18,17,0
91	2.47588E+27	90,83,82,0
92	4.95176E+27	91,12,11,0
93	9.90352E+27	92,1
94	1.9807E+28	93,20
95	3.96141E+28	94,10
96	7.92282E+28	95,48,46,1
97	1.58456E+29	96,5
98	3.16913E+29	97,10
99	6.33825E+29	98,46,44,1
100	1.26765E+30	99,38
101	2.5353E+30	100,6,5,0
102	5.0706E+30	101,66,65,0
103	1.01412E+31	102,8
104	2.02824E+31	103,10,9,0
105	4.05648E+31	104,15
106	8.11296E+31	105,14
107	1.62259E+32	106,64,62,1
108	3.24519E+32	107,30
109	6.49037E+32	108,6,5,0
110	1.29807E+33	109,12,11,0
111	2.59615E+33	110,9
112	5.1923E+33	111,44,42,1
113	1.03846E+34	112,8
114	2.07692E+34	113,81,80,0
115	4.15384E+34	114,14,13,0

ตารางที่ ก.1 (ต่อ)

116	8.30767E+34	115,70,69,0
117	1.66153E+35	116,19,17,1
118	3.32307E+35	117,32
119	6.64614E+35	118,7
120	1.32923E+36	119,117,110,6
121	2.65846E+36	120,17
122	5.31691E+36	121,59,58,0
123	1.06338E+37	122,1
124	2.12676E+37	123,36
125	4.25353E+37	124,107,106,0
126	8.50706E+37	125,36,35,0
127	1.70141E+38	126,0
128	3.40282E+38	127,28,26,1
129	6.80565E+38	128,4
130	1.36113E+39	129,2
131	2.72226E+39	130,47,46,0
132	5.44452E+39	131,28
133	1.0889E+40	132,51,50,1
134	2.17781E+40	133,56
135	4.35561E+40	134,10
136	8.71123E+40	135,125,124,0
137	1.74225E+41	136,20
138	3.48449E+41	137,7,6,0
139	6.96898E+41	138,7,4,2
140	1.3938E+42	139,28
141	2.78759E+42	140,31,30,0
142	5.57519E+42	141,20
143	1.11504E+43	142,20,19,0
144	2.23007E+43	143,69,68,0
145	4.46015E+43	144,51

ตารางที่ ก.1 (ต่อ)

146	8.9203E+43	145,59,58,0
147	1.78406E+44	146,37,36,0
148	3.56812E+44	147,26
149	7.13624E+44	148,109,108,0
150	1.42725E+45	149,52
151	2.8545E+45	150,2
152	5.70899E+45	151,65,64,0
153	1.1418E+46	152,0
154	2.2836E+46	153,128,126,1
155	4.56719E+46	154,31,30,0
156	9.13439E+46	155,115,114,0
157	1.82688E+47	156,26,25,0
158	3.65375E+47	157,26,25,0
159	7.30751E+47	158,30
160	1.4615E+48	159,18,17,0
161	2.923E+48	160,17
162	5.84601E+48	161,87,86,0
163	1.1692E+49	162,59,58,0
164	2.3384E+49	163,13,12,0
165	4.67681E+49	164,30,29,0
166	9.35361E+49	165,38,37,0
167	1.87072E+50	166,5
168	3.74144E+50	167,16,14,1

ภาคผนวก ข.
คุณสมบัติของเอฟพีจีเอตระกูล

คุณสมบัติของเอฟพีจีเอตระกูล

- Very low cost, high-performance logic solution for high-volume, consumer-oriented applications
  - Densities as high as 74,880 logic cells
  - Three power rails: for core (1.2V), I/Os (1.2V to 3.3V), and auxiliary purposes (2.5V)
- SelectIO™ signaling
  - Up to 784 I/O pins
  - 622 Mb/s data transfer rate per I/O
  - 18 single-ended signal standards
  - 6 differential I/O standards including LVDS, RSDS
  - Termination by Digitally Controlled Impedance
  - Signal swing ranging from 1.14V to 3.45V
  - Double Data Rate (DDR) support
- Logic resources
  - Abundant logic cells with shift register capability
  - Wide multiplexers
  - Fast look-ahead carry logic
  - Dedicated 18 x 18 multipliers
  - JTAG logic compatible with IEEE 1149.1/1532
- SelectRAM™ hierarchical memory
  - Up to 1,872 Kbits of total block RAM
  - Up to 520 Kbits of total distributed RAM
- Digital Clock Manager (up to four DCMs)
  - Clock skew elimination
  - Frequency synthesis
  - High resolution phase shifting

- Eight global clock lines and abundant routing
- Fully supported by Xilinx ISE development system
  - Synthesis, mapping, placement and routing
- MicroBlaze™ processor, PCI, and other cores
- Pb-free packaging options
- Low-power Spartan-3L Family and Automotive Spartan-3 XA Family options

Device	System Gates	Equivalent Logic Cells	CLB Array (One CLB = Four Slices)			Distributed RAM (bits[1])	Block RAM (bits[1])	Dedicated Multipliers	DCMs	Maximum User I/O	Maximum Differential I/O Pairs
			Rows	Columns	Total CLBs						
XC3S50[2]	50K	1,728	16	12	192	12K	72K	4	2	124	56
XC3S200[2]	200K	4,320	24	20	480	30K	216K	12	4	173	76
XC3S400[2]	400K	8,064	32	28	896	56K	288K	16	4	264	116
XC3S1000[2, 3]	1M	17,280	48	40	1,920	120K	432K	24	4	391	175
XC3S1500[3]	1.5M	29,952	64	52	3,328	208K	576K	32	4	487	221
XC3S2000	2M	46,080	80	64	5,120	320K	720K	40	4	565	270
XC3S4000[3]	4M	62,208	96	72	6,912	432K	1,728K	96	4	712	312
XC3S5000	5M	74,880	104	80	8,320	520K	1,872K	104	4	784	344

รูปที่ ข.1 แสดงรายละเอียดอุปกรณ์ในเอฟพีจีเอตระกูล Spartan 3